

CHRISTOPHER OCHOA

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EDUCATION

University of Oklahoma, Norman, OK

M.S. Aerospace Engineering, propulsion focus

Expected May 2027

B.S. Aerospace Engineering | GPA: 3.51

May 2026

RESEARCH EXPERIENCE

SOONER ADVANCED MANUFACTURING LAB, Norman, OK

2025 - Present

Graduate Researcher

- Lead preprocess modeling for four GE metal LPBF systems, including CAD preparation, build layout, FEA, support development, and slicing.
- Lead build setup, operation, and depacking for two GE MLab systems processing Inconel 718 and Ti-64
- Support a DoD contract involving CUI focused on additive manufacturing for aerospace repair applications.
- Conduct thesis research on hybrid LPBF and DED repair, characterizing interface behavior through process development and mechanical testing.
- Develop and evaluate metal AM processes through material characterization, parameter development, and process qualification.

UNIVERSITY OF OKLAHOMA, Norman, OK

2024 - 2025

Undergraduate Researcher

- Conducted funded research on regenerative-cooled rocket nozzles, using CFD and structural simulations to evaluate lattice-cooled designs.

PROJECTS

ADDITIVELY MANUFACTURED ROCKET NOZZLE

2025

- Designed, analyzed and printed a Ti-6Al-4V nozzle with body-centered cubic lattice cooling channels.
- Used FEA and build-strategy trade studies to assess thermal behavior, structural integrity and printability.

LEADERSHIP AND COMMUNITY INVOLVEMENT

DESIGN BUILD FLY TEAM, Norman, OK

2024 - Present

Structures Lead

- Lead aircraft structural design, analysis and fabrication, coordinating CAD, FEA and composite layup for a competition RC aircraft.

Other: President's Community Scholars; Oklahoma Ducks Unlimited; Boomer Rocket Team

TECHNICAL SKILLS

Additive manufacturing: LPBF, build preparation and process development; GE Concept Laser M2 & MLab

Design and analysis: SolidWorks, ANSYS, FEA, CFD, Python, C++, Pan Optimization FEA, nTopology

Materials: Inconel 718, 17-4 PH stainless steel, Ti-6Al-4V.

Fabrication: MIG welding, CNC milling, CNC plasma cutting, EDM, composite layup.

Languages: English (professional proficiency); Spanish (limited working proficiency).

HONORS AND AWARDS

- Eagle Scout, Troop 7
- International Baccalaureate Diploma (2021)